Setting Up an Interactive As-Built Floor Plan System

A step-by-step technical guide: clickable floor plans linked to room inventory records and 360° photography — for facility information management (PIM) use cases such as the Federal Secretariat, Lafia.

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1. What You're Building

The system has four layers that work together: a **digitized floor plan** where every room is a separately selectable shape, a **database of room records** (description, tags, inventory, documents), **360° photography** captured on site for each room, and a **viewer application** that ties the three together — click a room, see its record and panorama.

This is the same pattern used by commercial Property/Facility Information Management (PIM) platforms such as ActivePlan, Matterport, HoloBuilder, and OpenSpace. You can either license one of these, or build a lighter version yourself using open-source components. Both paths are covered in this guide.

Layer	Function	Typical Tools
Floor plan geometry	Vector shapes per room, each with a stable ID	SVG, GeoJSON, DXF/DWG export
Room data store	One record per room: metadata, tags, inventory	PostgreSQL, Firebase, Airtable
360° capture	Panoramic photo per room, tied to room ID	Ricoh Theta, Insta360, Matterport
Viewer application	Web app: click plan → load record → render panorama	React/HTML + Pannellum/Photo Sphere Viewer

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2. Hardware You'll Need

Only the photography step requires physical hardware. Everything else is software.

Purpose	Recommended Equipment	Notes
360° room capture	Ricoh Theta X / Z1, or Insta360 X3	Handheld, fast; good for single-room shots
3D + auto-dimensioning	Matterport Pro3	Adds LiDAR scanning; higher cost, richer output
Stabilisation	Lightweight tripod with levelling head	Keeps the horizon straight across all rooms
Field tagging	Tablet or laptop	Log which photo belongs to which room ID as you shoot

Purpose	Recommended Equipment	Notes
Dimension verification	Laser distance meter	Optional; cross-checks as-built vs. original CAD
Exterior/site context	Consumer drone	Optional; only if exterior views are in scope

Budget note: a Ricoh Theta or Insta360 camera plus a tripod is enough to pilot one floor for well under the cost of a Matterport Pro3 rig. Start there, and only invest in LiDAR capture if the project needs true 3D dollhouse views rather than flat 360° photos.

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3. Decide: License a Platform, or Build Custom

Option A — License an existing PIM/digital-twin platform

Fastest route. These platforms already combine plan, hotspots, 360° capture, and metadata.

- **Matterport** — best if you also want true 3D walkthroughs and auto-generated floor plans.
- **ActivePlan / similar PIM tools** — purpose-built for facility management with tagged inventory, closer to what's shown in your reference screenshots.
- **HoloBuilder / OpenSpace** — construction-progress focused, strong 360° timeline features.

Best when: you need to roll this out across many buildings quickly, don't have in-house developers to maintain custom software, and the recurring licence cost is acceptable against build/maintenance cost.

Option B — Build a custom lightweight system

Full control, works directly with your own CAD drawings and data, no recurring licence fee.

- **Floor plan rendering:** SVG in the browser, one shape per room with a unique ID.
- **360° viewer:** Pannellum or Photo Sphere Viewer — both free, open-source JavaScript libraries.
- **Room data store:** PostgreSQL or Firebase for production; Airtable is a good no-code starting point.
- **Frontend app:** React (or plain HTML/JS as in the working prototype already shared).
- **Image hosting:** S3 or Cloudinary — 360° photos are large and need a CDN.
- **Admin/CMS:** a simple form so facilities staff can edit inventory without touching code.

Best when: you need tight integration with other government systems, want to avoid recurring SaaS fees at national scale, or the platforms in Option A don't fit your data/workflow requirements.

4. Step-by-Step: Building the Custom Version

Step 1 — Get the source drawing in vector form

Request the original CAD file (DWG/DXF) from the architect rather than working only from the PDF. Vector geometry is far faster to convert into clickable shapes than a scanned or flattened drawing. If only a PDF is available, plan on manually tracing room outlines instead of an automated export.

Step 2 — Trace or export room polygons

Each room becomes its own closed shape (rectangle or polygon) with a unique, stable ID that matches your room-tag numbering (e.g. 00-002). Two ways to do this:

• From CAD: export room boundaries directly to SVG using AutoCAD's SVG export, or a script with the Python library `ezdxf`.

• From a scanned PDF: trace room outlines by hand in Inkscape, Illustrator, or Figma, saving as SVG with one path per room.

Step 3 — Assign room IDs consistently

Use one numbering scheme end-to-end: the SVG shape ID, the database record key, and the photo filename should all reference the same room tag (e.g. shape id="00-002", database key "00-002", photo "00-002.jpg"). This single convention is what makes the whole system click together.

Step 4 — Design the room data model

Create one record per room with the fields you actually need. A minimal schema:

```
room_id       text primary key   -- e.g. '00-002'
name          text               -- 'Banking Hall'
floor         text               -- 'Ground Floor'
description   text
tags          text[]             -- ['Public-facing','Furnished']
inventory     jsonb              -- list of items in the room
documents     text[]             -- linked file URLs
panorama_url  text               -- 360 photo location
updated_at    timestamp
```

Step 5 — Capture the 360° photography

Walk the building with the 360° camera, shooting one panorama per room from roughly the room's centre. Log each shot against its room ID on your tablet/laptop as you go — this tagging step is the one part of the whole workflow that cannot be automated later.

Step 6 — Upload photos and populate the database

Upload each panorama to cloud storage (S3/Cloudinary), named or keyed by room ID, and insert/update the matching database record with the resulting URL plus the inventory and description data for that room.

Step 7 — Build the viewer application

A minimal viewer needs three parts, wired together by room ID:

- Render the SVG floor plan in the page.
- On click of a room shape, fetch that room's record from the database/API.
- Populate a side panel with the record's fields and initialise a Pannellum/Photo Sphere Viewer instance pointed at `panorama_url`.

```
document.querySelectorAll('.room').forEach(el => {
  el.addEventListener('click', async () => {
    const record = await fetch(`/api/rooms/${el.id}`).then(r => r.json());
    renderPanel(record);
    pannellum.viewer('pano', {
      type: 'equirectangular',
      panorama: record.panorama_url
    });
  });
});
```

Step 8 — Add an editing interface

Give facilities staff a simple form (or an Airtable base, if you're not ready to build custom admin screens) to update descriptions, tags, and inventory per room without needing a developer.

Step 9 — Pilot on one floor, then scale

Validate the full workflow — capture, tagging, upload, viewing — on a single floor before committing to a full-building or multi-site rollout. This surfaces process issues (numbering mismatches, photo quality, missing rooms) while the fix cost is still low.

5. Suggested Rollout Plan

Phase	Activities	Output
1. Source data	Obtain CAD/DWG files; confirm room-tag numbering with the architect	Vector floor plan ready for tracing
2. Pilot capture	Shoot 360° photos for one floor; tag by room ID	Photo set for pilot floor
3. Prototype viewer	Build/deploy the SVG + panel + panorama viewer for the pilot floor	Working demo for stakeholder review
4. Data entry	Populate room records: description, tags, inventory	Populated database for pilot floor
5. Stakeholder review	Walk FPAM/client through the pilot; collect feedback	Sign-off or revision list
6. Full rollout	Repeat capture + data entry per floor/building; decide build vs buy at scale	System covering full site(s)

6. Practical Notes

- **Numbering discipline matters more than tooling.** Most real-world failures in these systems come from inconsistent room IDs between the drawing, the database, and the photo filenames — not from the software itself.
- **Vector source beats scanned PDF.** If a CAD file genuinely cannot be obtained, budget extra time for manual tracing; it is slower but produces the same end result.
- **Pilot before you scale.** One floor is enough to validate the capture-to-viewer workflow before committing camera time and developer time across an entire estate.
- **Licensing vs. building is a cost-over-time decision.** A licensed platform has lower upfront effort; a custom build has lower long-run cost at large scale and full data ownership.

Companion working prototype: interactive-floorplan-demo.html (SVG floor plan + room panel + 360° viewer, using Pannellum). Use it as a reference implementation for Step 7 above.